Predavanja

ustni izpit: 40%

predstaviteev: 20% (20 minut po 3 osebe)

Vaje

Projekt 40%

Logic in computer science

- Correctness of software/hardware/systems
  (watch Xavier Leroy: In search of sofware prfection)
- Developing proof asistants

Propositional logic (natural deduction proof system)
Simple type theory : Curry-Howard correspondance
(propositions as types)

Propositional logic

propositional formulas ϕ, ψ, θ

propositional variables p

atomic formula:

$$\phi ::= p \mid \psi \wedge \theta \mid \psi \vee \theta \mid \psi \to \theta \mid \neg\theta \mid \bot$$

TAVTOLOGIJA - validna formula (vedno res)

Natural deduction: proof system. Using formal rules we prove a formula ϕ

We can prove formula $\Leftrightarrow$ it is a tavtology

Primer: Proof of $p \wedge q \to p \vee r$

1. Assume $p \wedge q$ true (assumption)
2. p ($\wedge$elimination$_1$)
3. $p \vee r$ ($\vee$introduction$_1$)
4. $p \wedge q \to p \vee r$ ($\to$ introduction)

Za konstruktivno logiko (Brez LEM

$\wedge$

$$\frac{\phi \quad \psi}{\phi \wedge \psi}\wedge i \qquad \frac{\phi \wedge \psi}{\phi}\wedge e_1 \qquad \frac{\phi \wedge \psi}{\psi}\wedge e_2$$

premises

conclusion

$\vee$

$$\frac{\phi}{\phi \vee \psi}\vee i_1 \qquad \frac{\psi}{\phi \vee \psi}\vee i_2 \qquad \frac{\phi \vee \psi \quad \boxed{\begin{matrix}\phi\\ \vdots\\ \theta\end{matrix}} \quad \boxed{\begin{matrix}\psi\\ \vdots\\ \theta\end{matrix}}}{\theta}\vee e$$

$\rightarrow$

$$\frac{\boxed{\begin{matrix}\phi\\ \vdots\\ \psi\end{matrix}}}{\phi \rightarrow \psi}$$

box discharges the assumption ϕ

$$\frac{\phi \rightarrow \psi \quad \phi}{\psi}\rightarrow e$$

modus ponens

$\neg$

$\neg\phi \equiv \phi \rightarrow \bot$

$$\frac{\boxed{\begin{matrix}\phi\\ \vdots\\ \bot\end{matrix}}}{\neg\phi}\neg i \qquad \frac{\neg\phi \quad \phi}{\bot}\neg e$$

$\bot$

$$\frac{\bot}{\phi}\bot e$$

NO FORMUA IN THE BOX gets used outside the box.

Primer $p \wedge (q \vee r) \rightarrow (p \wedge q) \vee (p \wedge r)$

1. $p \wedge (q \vee r)$ assumption
2. p
3. $q \vee r$
4. q assumption
5. $p \wedge q$
6. $(p \wedge q) \vee (p \wedge r)$
7. r assumption
8. $p \wedge r$
9. $(p \wedge q) \vee (p \wedge r)$ $\vee e$ (3) (4-6) (7-9)
10. $(p \wedge q) \vee (p \wedge r)$ $\rightarrow i$ (1-10)

In classical logic

$$\frac{\boxed{\begin{matrix}\neg\phi\\ \vdots\\ \bot\end{matrix}}}{\phi}raa$$

Reductio ad absurdum

DN: prove LEM with raa

$$\frac{}{\phi \vee \neg\phi} \qquad \frac{\dfrac{\dfrac{\dfrac{}{\neg\phi}}{\bot}}{\phi}}{\phi \vee \neg\phi} \qquad \frac{\dfrac{}{\neg\phi}}{\phi \vee \neg\phi}$$

$$\phi \vee \neg\phi$$

A proof of formula ϕ from assumptions Γ is given by sequence of formulas following the rules such that all open assumptions in the proof are from Γ

notation: $\vdash \phi$

Theorem: $\vdash \phi$ (including raa) $\overset{\text{soundnes}}{\underset{\text{completness}}{\Longleftrightarrow}}$ ϕ is tautology

$\Leftrightarrow$ ϕ is true under any assigment of truth validnes to propositional variables that make all formulas in Γ true

Formulas $\phi ::= p \mid \phi \wedge \phi \mid \phi \vee \phi \mid \phi \to \phi \mid \bot$

Types $A ::= \alpha \mid A \times B \mid A + B \mid A \to B \mid$

$$t : A$$

term type

$$\frac{s : A \quad t : B}{(s, t) : A \times B} \qquad \frac{t : A \times B}{pr_1(t) : A}\times e_1 \qquad \frac{t : A \times B}{pr_2(t) : B}$$

variable

$$\frac{\begin{array}{c} x : A \\ \vdots \\ t : B \end{array}}{\lambda x : A . t : A \to B} \to i \qquad \frac{t : A \to B \quad u : A}{t u : B} \to e$$

term

$$\frac{t : A}{in_1(t) : A + B} \qquad \frac{t : B}{in_2(t) : A + B}$$

$$\frac{s : A + B \quad \boxed{\begin{array}{c} x : A \\ \vdots \\ t : C \end{array}} \quad \boxed{\begin{array}{c} y : B \\ \vdots \\ u : C \end{array}}}{\text{case } s \, (in_1(x : A) \mapsto t \mid in_2(y : B) \mapsto u) : C}$$

Primer:

A derived term of type $A \times B \to B \times A$

1.	$x : A \times B$	assumption	
2.	$pr_1(x) : A$	$\times e_1$	(1)
3.	$pr_2(x) : B$	$\times e_2$	(1)
4.	$(pr_2(x), pr_1(x)) : B \times A$	$\times i$	(3) (2)
5.	$\lambda x : A \times B . (pr_2(x), pr_1(x)) : A \times B \to B \times A$	$\to i$	(1-4)

tega ne potrebujemo

For every type we have introduction rules, eliminatio rules and computation rules

computation rule for rewrite rule (longer arrow)

$\times$: $pr_1(s,t) \longrightarrow s$

$pr_2(s,t) \longrightarrow t$

β-reduction

substitute u for x in t

$\rightarrow$: $(\lambda x{:}A.t)u \longrightarrow t[x:=u]$

(before doing this first rename bound variable in t that are distinct from variables in u)

$+$: $\text{case}\ (in_1(s))\ (in_1(x) \mapsto t \mid in_2(y) \mapsto u) \longrightarrow t[x:=s]$

0 : $\dfrac{t:0}{\text{empty}(t):A}$

DM: Find canonical term of type

$A \times (B+C) \longrightarrow A\times B + A \times C$